SCADA 2026 / FIELD-TO-POLICY REFLECTION

From AgTech to community-led climate adaptation

What needs to happen for agricultural technology to work for Indonesian smallholders?

By: Vania Sandra Clarissa

Why Does This Matter to Indonesia?

One important starting point is the structure of Indonesian agriculture.

BPS / AGRICULTURAL CENSUS 2023

17.25M

farmers operate on less than 0.5 hectares

62.05% of farmers using agricultural land



17,251,432 of 27,802,434 agricultural land users

This matters because small and fragmented landholdings create economic challenges.

WORLD BANK / 2030 MODELLED PROJECTIONS

Projected yield change without countermeasures

Maize	−7.1%
Irrigated rice	−0.72%

These are modelled projections, not observed national losses.

THE DESIGN CHALLENGE

How can Indonesia understand which technology solves which problem, who can operate it, and whether farmers can benefit economically from it

Sources: BPS, Sensus Pertanian 2023; World Bank, Indonesia Climate Resilient and Low Carbon Agriculture Development Project (P180681), 2023.

What Can Technology Actually Do?

The value is not the sensor itself. It is the decision the farmer can make with the information.

01 REMOTE SENSING

See the wider pattern

Satellite and drone data can flag vegetation stress, land-cover change and areas that need attention.

WHERE should we investigate?

02 PROXIMAL SENSING

Check the field

Soil and plant measurements help verify moisture, pH, nutrients and crop condition on the ground.

WHAT is happening on this plot?

03 PRECISION ACTION

Choose a response

Use the evidence to adjust irrigation or nutrient application through targeted management.

WHAT should change next?

+27%

higher agronomic efficiency

EAST LOMBOK / 2025 RICE TRIAL

Sensor-based fertilisation achieved 27% higher agronomic efficiency than conventional fertilisation under the study conditions.

The main lesson is that technology can help reduce guesswork and support more informed agricultural decisions.

Sources: Ismiani et al. (2025), Jurnal Agro 12(1), DOI: 10.15575/j.agro.44835; Indarto et al. (2022), Agriculture and Natural Resources 56(2).

The missing variable is adoption capacity.

A technology can be technically sound and still fail if people cannot test, trust, maintain or afford it.

THE EXISTING PLATFORM

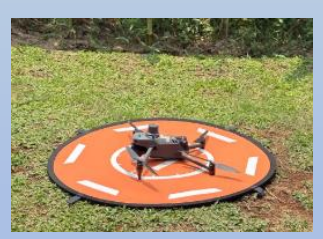
KELOMPOK TANI

Formed by, from and for farmers. A place for learning, collaboration and information exchange.











Who will operate it? Who will maintain it when something breaks? Who will interpret the data? And who will pay for the service once the project funding ends?

THE ADAPTABILITY PHASE

01 — LISTEN & MAP

Farmer priorities, soil, crop calendar, skills and buyer needs.

02 — TEST A DEMO PLOT

Compare the proposed practice with current methods over a crop cycle.

03 — TRAIN & ADAPT

Build local operation, maintenance and farmer feedback.

04 — REVIEW & DECIDE

Check cost, yield, usability and whether the service can continue.

Sources: Republic of Indonesia, Law No. 19/2013, Arts. 69–75; FAO, Farmer Field School approach.

The Economic Dilemma: Soil Health vs Farmers' Livelihoods

improving soil health does not automatically guarantee higher farmer income.

RURAL / OPEN FIELD

Restore and manage

Soil-health practices, agroforestry where suitable, field sensing and targeted input application.

Priority: resilience + input efficiency

PERI-URBAN / GREENHOUSE

Control key conditions

Drip fertigation and climate sensors for high-value horticulture close to urban markets.

Priority: productive land + reliable supply

URBAN / PFAL

Produce in controlled space

Hydroponics and artificial lighting for selected, fast-turnover crops; energy and capital costs matter.

Priority: supply consistency + energy fit

2 / 83

chili farmers in an IPB study used smart fertigation

Only 2.4% of the 83 farmers surveyed.

WHAT THE STUDY ALSO FOUND

Both adopters had longer training; all belonged to farmer groups.

The study observed higher productivity and profit among smart-fertigation farmers, despite higher farm costs. With only two adopters, this is a promising signal, not proof of a guaranteed return.

SOC recovery supports soil function over time; track yield and net income during the transition.

Sources: Tinaprilla et al. (2024), Jurnal Manajemen dan Agribisnis 21(1), DOI: 10.17358/jma.21.1.95; Minasny et al. (2018), European Journal of Soil Science.

Scale evidence, not equipment.

A five-year pathway should build local capability before expanding the technology footprint.

YEAR 1

01

UNDERSTAND

Map local soil, crops, climate risks, farmer priorities, group capacity and potential buyers.

YEARS 2–3

02

TEST & ADAPT

Run demonstration plots, train local operators, track full-season costs and results.

YEARS 4–5

03

SCALE WHAT WORKS

Secure service financing, buyer partnerships and monitoring before expansion.

CONTINUOUS MRV

Remote data + Field samples + Farmer records → Better decisions + course correction

Track yield, net income, input use, soil condition, water, GHGs, continued adoption and market access.

“The goal is not to deliver more technology. It is to leave farmers more capable of making good decisions without us.”

MRV should combine satellite/drone observations with field sampling, farm records and appropriate protocols; verified climate outcomes do not automatically create carbon-credit revenue.